

PALLAVI KOCHAR

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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Jan 2025 – May 2026

Master of Science in Finance (Concentration: Quantitative Finance & Data Analytics)

GPA: 3.9/4.0

- Research Assistant – DeFi Crypto Asset Pricing | Course Assistant – Financial Economics | MSF Program Ambassador
- Coursework: Financial Risk Management, Advanced Financial Derivatives, Quantitative Finance, Machine Learning

Indian Institute of Technology (IIT), Bombay

Jun 2018 – May 2022

Bachelor of Technology in Chemical Engineering (Minor: Industrial Engineering & Operations Research)

- Coursework: Probability & Stochastic Processes, Numerical Analysis, Differential Equations, Calculus, C++, Economics

PROFESSIONAL EXPERIENCE

fAlpha.ai

New York City, NY

Quant Data Engineer Intern

Mar 2026 – May 2026

- Engineered a SHAP-based feature attribution framework in Python to decompose multi-factor predictive equity models into per-feature sensitivity metrics; parallelized matrix operations to balance attribution fidelity against runtime
- Built an LLM-powered synthesis layer compiling multi-model predictive signals into structured equity research briefs

Kotak Securities Limited

Mumbai, India

Software Developer (Promoted 3x, IT Champion Award FY 2023–24)

Jun 2022 – Dec 2024

- Built the SIP auto-debit execution path for Smallcase baskets in C#/.NET and SQL Server, with idempotent retry and daily reconciliation so duplicate instructions under at-least-once delivery could not double-debit
- Optimized 50+ relational database queries, joins, and stored procedures across 10M+ financial records, ensuring strict data integrity and cutting manual reporting overhead by 50+ hours weekly
- Distributed AWS data pipelines (S3, Glue, Step Functions, Lambda, Docker) to ingest and sync unstructured market and CRM data, improving analytics workflow efficiency by 30%
- Engineered a Kafka/SQS streaming pipeline cross-checking PAN records against IP data, flagging 10+ fraudulent accounts monthly under KYC compliance

KEY PROJECTS

Multi-Asset Exotic Derivatives Pricing Model

Apr 2026 – May 2026

- Simulated correlated GBM paths via Cholesky factorization of the implied correlation matrix under 50% barrier, quarterly autocall, and memory-coupon logic; resulting in a Monte Carlo standard error of \$0.74 on a \$978 price per \$1,000 face
- Reconciled payoff against a static decomposition (zero-coupon bond, short worst-of down-and-in put, autocall strip) and estimated delta and vega by bump-and-revalue to suppress finite-difference noise

GARCH Volatility Modeling & Tail Risk (VaR/ES) Framework

Apr 2026 – May 2026

- Derived and optimized the GARCH(1,1) log-likelihood with stationarity and positivity constraints through reparameterization; estimates reconciled to fGarch and tseries to 4 decimal places
- Backtested 95% Value at Risk (VaR) with Kupiec unconditional-coverage and Christoffersen independence tests, comparing exception counts and clustering across GARCH, historical-simulation, and delta-normal estimators

Alternative Data Trading Strategy & Ablation Testing

Feb 2026 – Mar 2026

- Built an alternative-data equity strategy on satellite-derived production signals for Permian Basin E&P names, with LLM agents confined to qualitative roles behind a deterministic Python core so P&L remained reproducible and auditable
- Pre-registered the ablation set before evaluating results to separate genuine signal from specification search; across 18 trades on 10 tickers, achieved an 83% hit rate and a 0.35 Sharpe ratio per trade

Multi-Agent Systematic Equity Research & Backtesting

Dec 2025 – Jan 2026

- Built a hybrid dense-retrieval pipeline (BAAI/bge-base-en-v1.5, HNSW index via Qdrant) over 83K+ SEC 10-K filing chunks; cross-encoder reranking (ms-marco-MiniLM-L-6-v2) with citation scoring; p90 warm-query latency ~400ms
- Orchestrated 16 specialized research agents across a 503-ticker S&P 500 universe; 10.4-year historical backtest of Investment Clock sector rotation yielded 16.8% CAGR, +3.2% alpha over SPY benchmark

Interest Rate & Prepayment Modeling | Busey Bank Practicum

Sep 2025 – Nov 2025

- Modeled conditional prepayment rate with XGBoost across borrower and loan-characteristic segments and optimized hyperparameters via randomized cross-validation, cutting training latency by 25% while maintaining predictive accuracy
- Diagnosed target leakage in the CPR construction (prior and query balances mechanically embedded in the label), rebuilt the feature set, and flagged an inflated segment R^2 that would not have replicated out-of-sample

SKILLS AND CERTIFICATIONS

- **Quantitative Finance:** Monte Carlo Simulation, Stochastic Modeling, GARCH, VaR, Expected Shortfall, MLE, Black-Scholes, Binomial Trees, Portfolio Optimization, Variance Reduction, Cross-Validation, Risk Modeling
- **Mathematical & ML Methods:** Probability & Stochastic Processes, Linear Algebra, Time Series Analysis, XGBoost, LightGBM, Neural Networks, Bayesian Optimization, Explainable AI (SHAP), Logistic Regression, Agentic AI, GenAI
- **Programming & Infrastructure:** Python (NumPy, Pandas, SciPy, PyTorch), C++, C#/.NET, SQL (SQL Server, PostgreSQL), R, AWS (S3, Lambda, Glue, ECS), Apache Kafka, Docker, Git, Bloomberg, Capital IQ
- **Certifications:** Finance and Quantitative Modeling | Machine Learning | IBM Data Science | LangChain